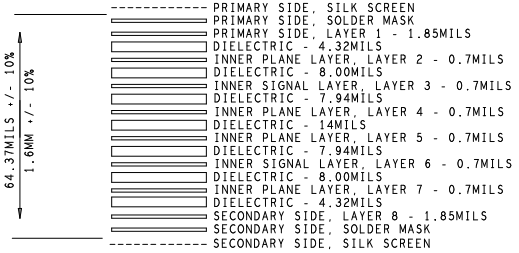
NOTES: UNLESS OTHERWISE SPECIFIED.

1. FABRICATE PER IPC-6012A CLASS 2.
2. MATERIAL:,
   DIELECTRIC: FR4 PER IPC-4101 WITH MIN TG 180 DEG C.
   COPPER: AS PER STACKUP.
   U.L. RATING: 94V-0 MINIMUM
3. SURFACE FINISH: EING.
4. SOLDER MASK MATERIAL SHALL MEET ALL THE REQUIREMENTS OF IPC-SM-840C AND SHALL BE GREEN IN COLOR AND APPLIED OVER BARE COPPER.
5. SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
6. 100% CONTINUITY TESTING USING DATABASE NETLIST SHALL BE PERFORMED, VENDOR TO IDENTIFY TEST PASSED IN SECONDARY SIDE.
7. VENDOR TO MARK DATE CODE AND LOGO IN ETCH OR IN LEGEND.
8. BOW AND TWIST SHALL NOT EXCEED 0.7% OF LONGEST SIDE .
9. DIELECTRICS AND LINEWIDTHS MAY BE ADJUSTED TO MEET THE IMPEDANCE REQUIREMENTS.
10. MINIMUM CONDUCTOR WIDTH: 4 MIL
    MINIMUM SPACING: 4 MIL
11. REQUEST FABRICATOR TO ADD TEAR DROPS IF NOT ADDED IN THE GERBERS.
12. PREFERRED LAMINATE/PREPREG FR4 OR FABRICATOR SHALL USE EQUIVALENT OF THE PREFERRED MATERIAL.
13. IF THE BOARD SIZE IS LESS THAN 50 X 50 MM, IT SHOULD BE IN A PANEL FORM FOR ASSEMBLY PURPOSE.
14. IF THE BOARD SIZE IS IRREGULAR (OTHER THAN SQUARE/RECTANGLE), EDGE PCB WITH BREAKAWAY TABS SHOULD BE THERE IN MINIMUM 2 SIDES WITH 5 MM HANDLING AREA.
15. KINDLY PROVIDE 5MM HANDLING AREA ON ALL SIDES, 4 MOUNTING HOLES AND 3 GLOBAL FIDUCIALS , BREAK AWAY TABS OR V-CUT TO BREAK PCB AFTER ASSEMBLY.
16. KINDLY NOTE 5 PEMNUT DETAILS, REFER DATASHEET & LOCATION OF DRILL SYMBOL [P]

CAUTION

SENSITIVE ELECTRONIC DEVICES-CLASS 1

LAYER STACK-UP



SINGLE ENDED

LAYER NO.	CONDUCTOR WIDTH	REFERENCE LAYER	IMPEDANCE +/-10%
LAYER 1	8.45 mil	2	50 OHM +/-10%
LAYER 3	8.15 mil	2 & 4	50 OHM +/-10%
LAYER 6	8.15 mil	5 & 7	50 OHM +/-10%
LAYER 8	8.45 mil	7	50 OHM +/-10%

DIFFERENTIAL PAIR

LAYER NO.	CONDUCTOR WIDTH	SPACING	REFERENCE LAYER	IMPEDANCE +/-10%
LAYER 1	5.50 mil	4.00 mil	2	90 OHMS+/-10%
LAYER 3	6.10 mil	4.00 mil	2 & 4	90 OHMS+/-10%
LAYER 6	6.10 mil	4.00 mil	5 & 7	90 OHMS+/-10%
LAYER 8	5.50 mil	4.00 mil	7	90 OHMS+/-10%

DIFFERENTIAL PAIR

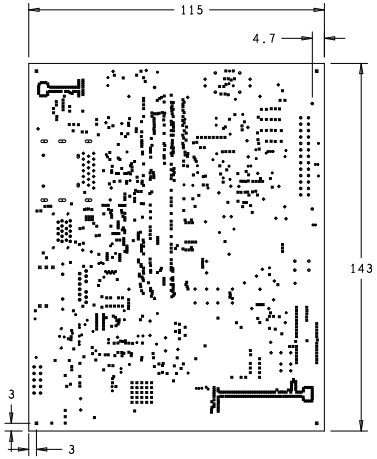
LAYER NO.	CONDUCTOR WIDTH	SPACING	REFERENCE LAYER	IMPEDANCE +/-10%
LAYER 1	4.00 mil	4.05 mil	2	100 OHMS+/-10%
LAYER 3	-	-	-	100 OHMS+/-10%
LAYER 6	4.55 mil	4.00 mil	5 & 7	100 OHMS+/-10%
LAYER 8	4.00 mil	4.05 mil	7	100 OHMS+/-10%

DIFFERENTIAL PAIR

LAYER NO.	CONDUCTOR WIDTH	GND SEPERATION	REFERENCE LAYER	IMPEDANCE +/-10%
LAYER 1	4.1 mil	4.8 mil	2	120 OHM +/-10%
LAYER 8	4.1 mil	4.8 mil	7	120 OHM +/-10%

COPLANAR-SINGLE ENDED

LAYER NO.	CONDUCTOR WIDTH	GND SEPERATION	REFERENCE LAYER	IMPEDANCE +/-10%
LAYER 1	20 mil	8.00 mil	3	50 OHM +/-10%



EMBITEL	
DATE:19-07-2021	JOB NUMBER:EMBL51745
DESIGNED AT:SIENNA ECAD TECHNOLOGIES	
PART NO/CARD REF:PCB_CB_V2.0	REV:2.0
FILM LAYER: FABRICATION LAYER	

DRILL CHART: TOP to BOTTOM

ALL UNITS ARE IN MILS

FIGURE	SIZE	TOLERANCE	PLATED	QTY
▪	10.0	+3.0/-3.0	PLATED	910
•	12.0	+3.0/-3.0	PLATED	98
•	23.62	+3.0/-3.0	PLATED	2
•	27.56	+3.0/-3.0	PLATED	18
•	35.04	+3.0/-3.0	PLATED	12
•	39.37	+3.0/-3.0	PLATED	84
•	40.16	+3.0/-3.0	PLATED	10
•	43.31	+3.0/-3.0	PLATED	28
•	47.24	+3.0/-3.0	PLATED	4
▪	50.0	+3.0/-3.0	PLATED	5
•	61.81	+3.0/-3.0	PLATED	2
•	90.55	+3.0/-3.0	PLATED	4
▪	125.98	+3.0/-3.0	PLATED	4
•	166.14	+3.0/-3.0	PLATED	5
▲	43.31	+2.0/-2.0	NON-PLATED	1
•	62.99	+2.0/-2.0	NON-PLATED	1
•	86.61	+2.0/-2.0	NON-PLATED	2
•	90.55	+2.0/-2.0	NON-PLATED	2
•	125.98	+2.0/-2.0	NON-PLATED	2
•	127.95	+2.0/-2.0	NON-PLATED	2
▪	62.99x35.43	+3.0/-3.0	PLATED	2
▭	102.36x39.37	+3.0/-3.0	PLATED	4
▭	110.24x39.37	+3.0/-3.0	PLATED	2

UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN ☐ INCHES ☒ MILLIMETERS

DATE 19/07/2021

SIENNA ECAD Technologies

DRAWING TITLE: FABRICATION DRAWING

SIZE D | PART NUMBER: CARRIER CARD | REV 2.0

SCALE 1/1 | SHEET 1 OF 1

CAD DATA 2 DO NOT MANUALLY 1 UPDATE